

Explaining the Matthew Effect using Exogenous Knowledge Demand Shocks: A Study of Biomedical Research using Epidemics as Natural Experiments

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1 Introduction and Theoretical Background

The complex pattern of the misallocation of credit for scientific work must quite evidently be described as “the Matthew effect,” for, as will be remembered, the Gospel According to St. Matthew puts it this way: For unto every one that hath shall be given, and he shall have abundance: but from him that hath not shall be taken away even that which he hath. Put in less stately language, the Matthew effect consists of the accruing of greater increments of recognition for particular scientific contributions to scientists of considerable repute and the withholding of such recognition from scientists who have not yet made their mark.

Merton Robert K.
The Matthew Effect in Science, 1968

While the Matthew effect ([Merton \(1968\)](#)) is already well documented, including in the specific case of science and academic research, it remains unclear to what extent the accumulation of scientific credit in the hands of a few is occurring either because of a supposed “misallocation” or a latent long-tailed distribution of scientific prowess. To disentangle the two, we propose in this research project to conceive science as a market where the demand for knowledge from society meets a supply of knowledge produced by scientists. Researchers choose which field of study to contribute to based on their own specialties and the prevailing demand for knowledge. They therefore make a trade-off between producing knowledge of higher quality and responding to society’s demand for knowledge (particularly in biomedical sciences). We assume that this trade-off is adjusted by citations, which act as a professional reward for a researcher’s work and career choices (depending on the scientific field, this role of citations is implicit or deliberate, with the *h-index*, for instance). Citing a researcher’s work rewards her both for the quality of her work as well as its relevance, since the citing researcher only references work she finds useful.¹

¹It is worth noting that securing funding is also an important objective in a researcher’s agenda ([Azoulay and Fishman \(2020\)](#)). However, we argue that funding is granted based on parameters that also depend on the researcher’s quality of work and the current demand for knowledge: being well published and well cited contribute to securing research funding.

We argue that the citation-based market adjustment mechanism is imperfect and subject to a number of frictions, and that these are at the source of the stylized fact we call the "Matthew effect". Such "friction" happens because the intrinsic quality of a scientific article is not directly accessible². The citing researcher (demand side) must make a choice based on a set of criteria (signals) that are correlated with, but not identical to, the quality of the articles she wishes to cite: the prestige of the researcher (e.g. their existing number of citations from other works), various social attributes (e.g. the researcher's network, gender, education), the visibility of the article (e.g. the journal of publication), and so on. Researchers can be considered rational individuals, faced with a simple time (paper quality investigating efforts) constraint. Using the general attributes of an article is for them a strategy to identify articles of interest without having to investigate too much its quality or the whole literature. Because they act as signals, we should expect that they play a bigger role in universes with higher uncertainty / scarcer information / higher costs to investigate quality.

In this research project, we propose a novel setting to evaluate the role played by an article's attributes (signals) in the decision-making process of a citing research. We use epidemics as external demand shocks, creating a rise in uncertainty in article quality because of (1) rising knowledge supply (more articles are being published in the field and there is a bigger literature to explore), (2) increased pace of research (there is the need to produce knowledge faster, and therefore less time to investigate or peer review papers), and (3) recomposition effects (non-specialists with less knowledge on the existing literature enter the research field). We focus on biomedical research because actors of this field explicitly rely on scientometric measures to identify the literature they want to contribute to (cite). This is justified by the fact that biomedical research has been facing in the last decades well-identified obstacles such as the difficulty in replicating results or a vast, fast-evolving international knowledge production.

Research Question: Do the increased uncertainty associated with external knowledge demand shocks modify the citation rationales in academic research?

H1: An increase in uncertainty leads to an increase in signal-related citation patterns. Specifically, the scientific prestige of a researcher (in terms of citations, career, or other social attributes) will have a stronger predictive effect on the number of citations when the research field is younger, is wider, or when the pace of research is faster.

H2: High-prestige researchers are more likely to transition between research fields than other researchers during periods of increased uncertainty because - for a fixed entry cost - they can maintain higher payoffs (measured as in number of citations).

Signal-related biases in the scientific citation mechanism are generally difficult to uncover because 1) there is no way to measure the "intrinsic quality" of an article, 2) the quality of an article and the aforementioned biases are highly correlated (e.g., highly cited researchers tend to produce higher-quality research), 3) changes in citation mechanisms within a scientific field often follow internal dynamics. For instance, a scientific innovation may create a research bubble with speculations about the most promising research directions. In this context, the "demand" for knowledge and the "quality" of research are entirely endogenously determined. It is therefore necessary to identify an experiment when the quality and propensity for innovation in research remain stable, but where the demand for knowledge and uncertainty about knowledge quality increase exogenously. Such "shocks to knowledge demand" have been observed during epidemic episodes. For coronavirus for instance, which will be our main empirical setting, we have SARS-

²In fact, most of the literature discussing performance in scientific research use the number of citations and the journal of publication's ranking as face value for the quality of a paper, a researcher, or an institution.

CoV (2002-2004), MERS-CoV (2012-2014) and SARS-CoV-2 (2019-2022).

The increase in demand (and hence the uncertainty regarding research quality) will be measured by the rise in citations for papers specific to coronaviruses during an epidemic. Counts of cases and/or deaths in the world and/or Western countries, with or without lags, will be used as instrumental variables to assess the robustness of results.

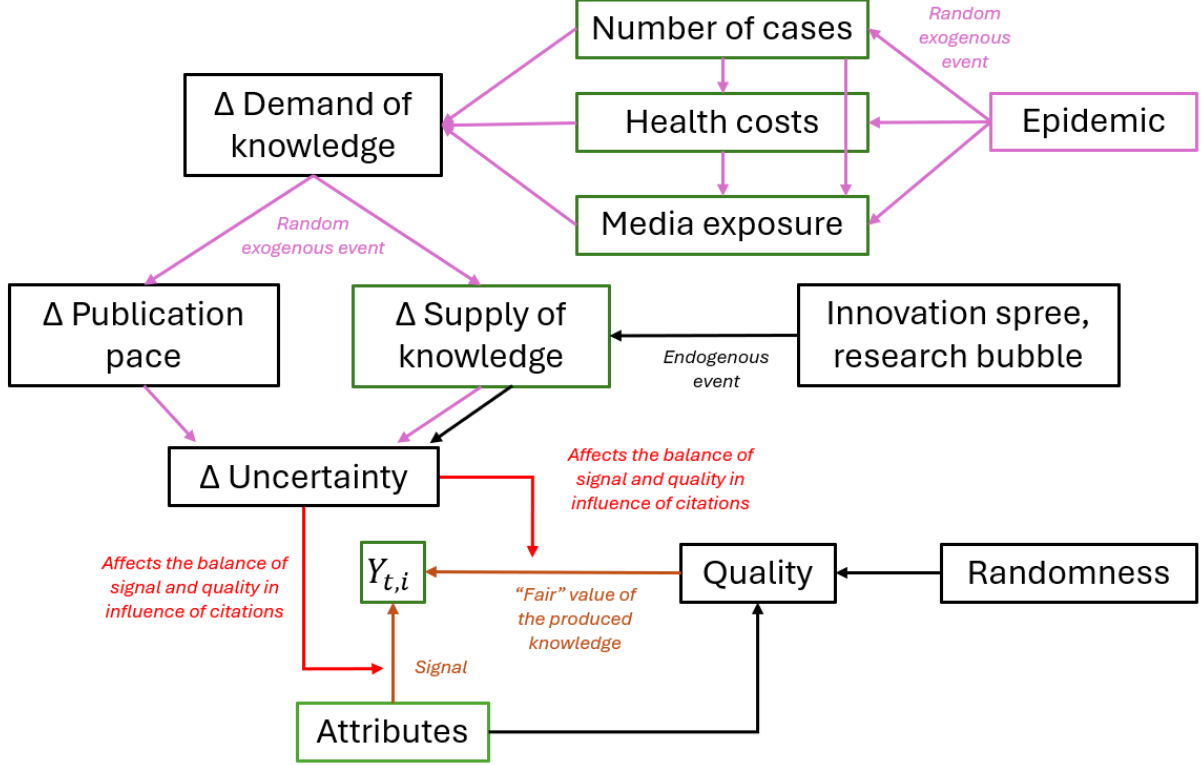


Figure 1: Theoretical framework with measurable (green) and hidden (black) variables, effects of interest (orange and red), instruments (pink).

Note: $Y_{t,i}$ can have different specifications, including the number of citations gained by article i in between dates $t-1$ and t ($\Delta Citations$), its market share ($\frac{\Delta Citations}{Demand}$), or a weighted market share ($\frac{Supply \times \Delta Citations}{Demand}$).

2 Literature

The empirical evidence for this research project draws on aspects of the general STEM literature, scientometrics, and sociology of science describing multiple aspects of the Matthew effect:

- Scientific career trajectories [Katchanov et al. \(2023\)](#)
- Gender biases in citation mechanisms [Squazzoni et al. \(2021\)](#) and editorial decisions [Card and DellaVigna \(2020\)](#).
- North-South inequalities in research visibility [INSERT KEY CITATION]
- Publication strategies linked to researcher collaboration [Larivière et al. \(2013\)](#)

- Favoritism in the supposedly meritocratic system of academic research [Fisman et al. \(2018\)](#), [Bao and Huang \(2024\)](#)

While such literature is prolific and provides much anecdotal evidence of the mechanisms described in the introduction, very little tries to disentangle the effect of impact on paper quality from its general attributes on expected academic success:

- [Azoulay et al. \(2014\)](#) and [Reschke et al. \(2018\)](#) use Howard Hugues Medical Institute investigator appointments as random events that signal (or not) the quality of the previous work done by a researcher.

We differentiate from such works by looking rather not at fluctuations in a researcher’s characteristics, but at fluctuations in her environment. What are the characteristics of a research field that allow the Matthew effect to take place? [Hill and Stein \(2025a\)](#) show, for instance, that increased competition in a research field leads to a lower quality of papers published in the field.

Our testing hypotheses and theoretical setting directly build on the literature of economics of information. The quality of a paper is not directly measured by agents (researchers) but is inferred through signals (general attributes of the paper):

- [Carnehl and Schneider \(2024\)](#) examined principal-agent mechanisms in the allocation of ERC funding and the consequences on the structure of research.
- [Card and DellaVigna \(2020\)](#) modeled editorial decision-making processes in four major American economic journals, showing that peer-review decisions align with the goal of maximizing future citations.
- [Hill and Stein \(2025b\)](#) demonstrated that being the first to publish a study has a significant impact on researchers’ careers and strongly influences researchers’ beliefs. They adapted Card’s model to one in which information is noisy.

Finally, to test H2, the project will also contribute to macroeconomics and applied industrial economics, as it involves optimizing the production of knowledge across multiple research domains under demand and productivity constraints:

- [Jones \(2021\)](#) documented how authors choose whether or not to work in teams to maximize shared citations.

3 Theoretical Model

3.1 Model for H1

Citing authors choose the article with the best quality. They do not observe it directly and therefore must make assumptions about the quality of the articles. Articles are free public goods, and competition only happens through quality. Citing authors choose only one article.

- q : Intrinsic quality of the article
- $\hat{q}$: Observed quality of the article
- $S \in [0, \infty[$: Observable characteristics of the article (authors, publication journal, etc.)
- $\epsilon \sim N(0, \sigma^2)$: Variation of the intrinsic quality of the article around the observable characteristics of its publication
- $\hat{\epsilon} \sim N(0, \hat{\sigma}^2)$: Noise around the intrinsic quality of the article
- $p \in [0, P]$: Effort exerted by the citing author to reveal q and minimize noise

We have:

$$\begin{aligned} q &= S + \epsilon \\ \hat{q} &= q + \frac{\hat{\epsilon}}{p} \end{aligned}$$

Thus, the observed quality is unbiased:

$$E(\hat{q}) = E(S + \epsilon + \frac{\hat{\epsilon}}{p}) = S$$

Suppose the citing author must arbitrate between citing an article i or j based on their respective quality. They reason probabilistically:

$$A : P(q_i > q_j) = P\left(\hat{q}_i - \frac{\hat{\epsilon}_i}{p_i} > \hat{q}_j - \frac{\hat{\epsilon}_j}{p_j}\right) = P\left(\hat{q}_i - \hat{q}_j > \frac{\hat{\epsilon}_i}{p_i} - \frac{\hat{\epsilon}_j}{p_j}\right)$$

And:

$$B : P(q_i > q_j) = P(S_i - S_j > \epsilon_i - \epsilon_j)$$

To make a decision, since there is no bias, the citing author chooses the calculation where they will be most certain of the final result. That is, the calculation where the estimator has the lowest variance:

$$\begin{aligned} A : \sigma^2(\epsilon_i - \epsilon_j) &= 2\sigma^2 \\ B : \sigma^2\left(\frac{\hat{\epsilon}_i}{p_i} - \frac{\hat{\epsilon}_j}{p_j}\right) &= \frac{\hat{\sigma}_i^2}{p_i^2} + \frac{\hat{\sigma}_j^2}{p_j^2} \end{aligned}$$

Assume that the sigmas are equal among themselves. This assumption is reasonable since the beliefs of the citing author are general to the field of research and not specific to each article:

$$2\sigma^2 < \frac{\hat{\sigma}^2}{p_i^2} + \frac{\hat{\sigma}^2}{p_j^2}$$

If generalized to n articles (recall that P is the total effort a citing author is willing to exert to choose their article):

$$\sum_{n \in [1, N]} p_n = P$$

$$N\sigma^2 < \sum_{n \in [1, N]} \frac{\hat{\sigma}^2}{p_n^2}$$

Since all sigmas are equal, the right-hand side of the inequality is minimized when all p_n are equal. Thus:

$$N\sigma^2 < \frac{N\hat{\sigma}^2}{\left(\frac{P}{N}\right)^2}$$

And:

$$P < N^2 \frac{\hat{\sigma}}{\sigma}$$

We thus formalize three phenomena:

1. The larger the research field (the more there are published articles, N), the more citing authors choose decision mode A . The relationship is quadratic, so one can expect a certain threshold beyond which S becomes a significantly stronger predictor of an article's success.
2. The greater the uncertainty about the quality of articles (e.g. when the field is young or research pace is increasing, $\hat{\sigma}$ or maybe even $1/P$), the more citing authors choose decision mode A .
3. The less reliable the authors' characteristics are in predicting an article's quality (e.g. when researchers are young or the field is new, σ), the more citing authors rely on B .

3.2 Model for H2

To validate H2, we can use the results of M1 and focus on the supply side of the market. Researchers aim to maximize their number of citations in $t + 1$ by choosing to conduct research in a field (a market) $m_1, m_2, \dots, m_n$. They make this choice based on their specializations (their production characteristics in the markets $m_1, m_2, \dots, m_n$) - which influences the quality of their work - as well their ability to produce a positive signal, and the demand for knowledge in each market. Let us define :

- A : Signal quality based on the researcher's characteristics
- $\epsilon \sim \mathcal{N}(0, \sigma^2)$: Variation in the expected quality of the article around its expected quality based on the authors' specializations
- $q = A + \epsilon$: Intrinsic quality of the article
- $\hat{\epsilon} \sim \mathcal{N}(0, \hat{\sigma}^2)$: Noise around the intrinsic quality of the article
- $\widehat{q}_m = q + \hat{\epsilon}$: Quality of the article as observed by potential citers in market m
- D_m : Demand in market m (number of citations that will happen in the field m)
- S_m : Supply in market m (number of papers published in the field m)
- U_m : Expected demand (utility) perceived by the researcher for his paper i if she produces in market m

We have:

$$E(\widehat{q_m}) = E(A + \epsilon + \widehat{\epsilon}) = A$$

According to the M1 modeling, agents have heterogeneous thought processes. They observe $\widehat{q_m}$ but aim to determine the true quality of the article q by forming a belief q_m^* . To conceptualize collective rather than individual rationality, we can combine the two modes of reasoning through a Bayesian model:

$$q_m^* = \alpha \times \widehat{q_m} + (1 - \alpha) \times A \quad \alpha \in [0, 1]$$

$$q_m^* = A + \alpha(\epsilon_i + \widehat{\epsilon}_i)$$

For researcher i and the set $[1, S_m]$ of researchers publishing in the field m , we have:

$$U_{m,i} = E(D_m \times \mathbb{I}[\max_{n \in [1, S_m]} q_{m,n}^* = q_{m,i}^*])$$

$$U_{m,i} = D_m \times P\left(\bigcap_{n \in [1, S_m], n \neq i} q_{m,i}^* > q_{m,n}^*\right)$$

$$U_{m,i} = D_m \times P\left(\bigcap_{n \in [1, S_m], n \neq i} \epsilon_n + \widehat{\epsilon}_n - \epsilon_i - \widehat{\epsilon}_i < \frac{A_i - A_n}{\alpha}\right)$$

Let $e = \epsilon + \widehat{\epsilon} \sim \mathcal{N}(0, (\sigma + \widehat{\sigma})^2)$:

$$U_{m,i} = D_m \times \int_{-\infty}^{+\infty} P\left(\bigcap_{n \in [1, S_m], n \neq i} e_n - x < \frac{A_i - A_n}{\alpha}\right) \times P(e_i = x) dx$$

$$U_{m,i} = D_m \times \int_{-\infty}^{+\infty} \prod_{n \in [1, S_m], n \neq i} P\left(e_n < \frac{A_n - A_i}{\alpha} + x\right) \times P(e_i = x) dx$$

For A_m the average value on n of A_n :

$$U_{m,i} = D_m \times \int_{-\infty}^{+\infty} \Phi\left(\frac{1}{\sigma + \widehat{\sigma}} \times \left(\frac{A_i - A_m}{\alpha} + x\right)\right)^{S_m-1} \times \phi(x) dx$$

Thus, we formalize three phenomena:

1. The expected utility for researcher i of entering the research field m decreases with the expertise of other researchers in the field (A_m).
2. The expected utility for researcher i of entering the research field m decreases with the number S_m of researchers in the field. This utility decreases even more rapidly when A_i is low.

3. The importance of A_m in the final decision-making decreases with its predictive capacity for the quality of the article (which varies inversely with σ).

However, there are issues of general equilibrium tbd

4 Empirical Strategy

4.1 Data

We study articles made available by the PubMed database. We do not rely on more extensive cross-country databases as we want to capture homogeneous demand markets, without any geographical or cultural barriers. We define a research field as the intersection between a "topic" (a type of disease) and an "approach" (which can be either considered as a research method, or an objective-driven framework). Articles are placed in this 2D space (although fields are not mutually exclusive) using keyword search in the title and abstract of papers:

Topic \ Approach (Number of identified articles)	All	epidemiology	diagnosis	treatment	prevention	(meta-analysis) OR (comparative study)	DNA OR RNA	vaccine	molecular biology	survey	clinical trial	simulation
shingles	434	83	132	305	163	7	11	195	2	26	18	4
chickenpox	987	217	280	488	250	24	26	240	4	53	24	6
mpox	1569	619	389	608	653	38	134	564	39	134	22	36
RSV	2557	569	737	1313	977	167	330	945	74	168	167	30
japanese encephalitis	3462	1025	1202	1213	1098	226	713	1251	98	224	130	42
polio	3770	1449	682	2452	1970	166	104	2245	28	588	197	27
mumps	3957	1111	1265	1703	1236	297	291	2207	62	271	210	18
norovirus	4002	1696	1003	1201	1121	194	1209	653	362	271	50	118
chikungunya	4073	1646	1473	1270	898	134	849	737	277	328	71	144
diphtheria	6337	950	1045	2679	1359	410	312	2556	134	240	508	44
meningococcal	6374	2497	2183	3589	2606	373	296	3294	83	601	556	45
ebola	6864	3072	1409	3813	3018	138	846	1471	303	558	363	274
rubella	7087	1886	2820	3120	2543	613	426	3760	89	716	306	38
zika	7669	2680	2313	2377	2108	129	1537	1472	652	485	77	270
tetanus	8578	1278	1863	4723	2321	627	148	4768	53	472	680	43
rotavirus	9879	3737	2854	3968	3493	880	2407	4465	491	909	580	114
rabies	10025	3051	2893	4962	4561	563	1101	5494	341	658	263	159
pneumococcal	13359	4566	4570	8293	6194	935	681	8564	261	1138	1177	155
measles	13675	5209	3627	7136	5758	921	1172	8221	235	1446	516	133
dengue	19234	6834	6914	5903	4551	819	3115	3482	1207	1693	523	665
coronavirus	31074	9909	9374	12916	7723	1290	3800	4459	1209	2953	749	741
pertussis OR (whooping cough)	31151	4718	5879	14862	8859	2410	2956	12680	875	1919	1551	232
HPV OR Papillomavirus	59590	16549	26435	26511	17893	4242	22616	17295	2744	7422	2940	600
influenza	73365	24293	19594	32918	26769	4589	9757	29597	2866	7758	3461	1877
hepatitis	138745	33553	55322	76340	25077	10185	42261	18138	3969	10847	8626	1175
virus	365307	57916	120754	114423	55909	24162	128272	51291	19782	13567	8067	4607
HIV OR AIDS	437295	124233	162427	254394	121520	24863	59815	31062	12518	51265	31048	7954

Figure 2: Number of identified articles in PubMed by topic / approach.

4.2 Empirical Strategy for H1

Let us define:

- $C_{t,i}$: Number of citations of article i at date t
- $\Delta C_{t+1,i} = C_{t+1,i} - C_{t,i}$: Number of citations gained by article i between date t and $t + 1$
- D_t : Knowledge demand (total number of citations given at time t)

- S_t : Knowledge supply (total number of articles published at time t)
- $M_{t,i} = \frac{\Delta C_{t,i}}{D_t}$: Market power of article i at date t
- $\widehat{M}_{t,i} = \frac{S_t \Delta C_{t,i}}{D_t}$: Scaled market power of article i at date t
- A_i : Attributes of the paper i , defined in level as the quality (mean number of citations) of the paper depending of these attributes
- A_t : Mean level of the attributes of the researchers active in the market at time t
- $\widehat{A}_{i,t} = \frac{A_i}{A_t}$
- q, s : Effects of quality and signal
- Q_i : Quality of the paper i

We measure A_i as the prediction of the following OLS³ where X_i is a vector of characteristics pertaining to article i (author characteristics, journal of publication, etc.) and α_t a time fixed effect to capture D_t and S_t . $Y_{t,i}$ can be replaced by $\Delta C_{t,i}$, $M_{t,i}$ or $\widehat{M}_{t,i}$ for alternative specifications:

$$\begin{aligned}
 (1) \quad Y_{t,i} &= \beta X_{i,t} + \alpha_t + res_i \\
 &\Rightarrow A_i = \beta X_i \\
 &\Rightarrow E(A_i) = C_i
 \end{aligned}$$

We want to measure q and s . If we could measure Q , we would implement the following OLS:

$$Y_{t,i} = q_t Q_i + s_t A_i + \epsilon_{t,i}$$

We make a hypothesis on the relationship between Q and A . The quality of the paper i is on average equal to A but with a random noise of mean 0 and unknown variance:

$$Q_i = A_i + r_i, \quad \mathcal{L}(0, \sigma_r)$$

Based on (1) we can impute:

$$\sigma_r = \sigma(res_i)$$

Thus:

$$\begin{aligned}
 Y_{t,i} &= q_t(A_i + r_i) + s_t A_i + \epsilon_{t,i} \\
 &\Rightarrow Y_{t,i} = (q_t + s_t)A_i + q_t r_i + \epsilon_{t,i}
 \end{aligned}$$

We can implement the following regression and estimate a coefficient β econometrically:

$$Y_{t,i} = \beta_t A_i + e_{t,i}$$

³We can also choose to implement a more complex algorithm such as a random forest or a neural network and define res_i as the error of the implemented model.

We have :

$$\begin{aligned}
(1) \quad & q_t + s_t = \beta_t \\
(2) \quad & q_t r_i + \epsilon_{t,i} = e_{t,i} \\
& \Rightarrow \sigma(q_t r_i + \epsilon_{t,i}) = \sigma(e_{t,i}) \\
& \Rightarrow q_t \sigma(r_i) + \sigma(\epsilon_{t,i}) = \sigma(e_{t,i}) \\
& \Rightarrow q_t \sigma_r + \sigma_{\epsilon_t} = \sigma_{e_t}
\end{aligned}$$

We make the assumption that $\sigma_{\epsilon_t} = \sigma_{\epsilon}$. ϵ_t could vary through time because of better transparency of the information, either pertaining to Q or to A . While this is not true in the long term (with the development of the internet, database collections systematically referencing ORCID numbers, h-indexes, article keywords, etc.), it may be reasonable to hypothesize that this is not the case in the short term:

$$\begin{aligned}
q_t \sigma_r + \sigma_{\epsilon} &= \sigma_{e_t} \\
\Rightarrow q_t &= \frac{\sigma_{e_t} - \sigma_{\epsilon}}{\sigma_r} \\
\Rightarrow s_t &= \beta_t - \frac{\sigma_{e_t} - \sigma_{\epsilon}}{\sigma_r}
\end{aligned}$$

We are interested rather not by s_t and q_t but by $s_t - q_t$ and its relationship with D . Since we know different levels of D_t for multiple t , we can look at:

$$\begin{aligned}
s(D) - q(D) &= \beta(D) - 2 \frac{\sigma_e(D) - \sigma_{\epsilon}}{\sigma_r} \\
\Rightarrow \frac{\partial(s - q)(D)}{\partial D} &= \frac{\partial \beta(D)}{\partial D} - 2 \sigma_r^{-1} \frac{\partial \sigma_e(D)}{\partial D}
\end{aligned}$$

As we can estimate $\beta(D)$, $\sigma_e(D)$ and σ_r , we can thus estimate the variation of different $(s - q)$ for different D and test H1.

4.3 Empirical Strategy for H2

ISSUES :

Empirical model does not account for Supply and Average Expertise

What about general equilibrium

Propose expertise as a proxy of quality or entry costs

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